Table 2

Tier Level	Requirements	Possible unavailability in a given year
1	• Single non-redundant distribution path serving the IT equipment • Non-redundant capacity components • Basic site infrastructure with expected availability of 99.671 per cent	1729.224 minutes (28.8 hours)
2	• Meets or exceeds all Tier 1 requirements • Redundant site infrastructure capacity components with expected availability of 99.741 per cent	1361.304 minutes (22.6 hours)
3	• Meets or exceeds all Tier 1 and Tier 2 requirements • Multiple independent distribution paths serving the IT equipment • All IT equipment must be dual-powered and fully compatible with the topology of a site's architecture • Concurrently maintainable site infrastructure with expected availability of 99.982 per cent	94.608 minutes (1.5 hours)
4	• Meets or exceeds all Tier 1, Tier 2 and Tier 3 requirements • All cooling equipment is independently dual-powered, including chillers, heaters, ventilation and air-conditioning (HVAC) systems • Fault-tolerant site infrastructure with electrical power storage and distribution facilities with expected availability of 99.995 per cent	26.28 minutes (0.4 hours)

Using cloud simulators, researchers can execute their algorithmic approaches on a software-based library and can get the results in different parameters including energy optimisation, security, integrity, confidentiality, bandwidth, power and many others.

Tasks performed by cloud simulators

The following tasks can be performed with the help of cloud simulators:

- Modelling and simulation of large scale cloud computing data centres
- Modelling and simulation of virtualised server hosts, with customisable policies for provisioning host resources to VMs
- Modelling and simulation of energy-aware computational resources
- Modelling and simulation of data centre network topologies and message-passing applications
- Modelling and simulation of federated clouds
- Dynamic insertion of simulation elements, stopping and resuming simulation
- User-defined policies for allocation of hosts to VMs, and policies for allotting host resources to VMs

Scope and features of cloud simulations

The scope and features of cloud simulations include:

- Data centres
- Load balancing
- Creation and execution of cloudlets
- Resource provisioning
- Scheduling of tasks
- Storage and cost factors
- Energy optimisation, and many others

Cloud simulation tools and plugins

Cloud simulation tools and plugins include:

- CloudSim
- CloudAnalyst
- GreenCloud
- iCanCloud
- MDCSim
- NetworkCloudSim
- VirtualCloud
- CloudMIG Xpress
- CloudAuction
- CloudReports
- RealCloudSim
- DynamicCloudSim
- WorkFlowSim

CloudSim

CloudSim is a famous simulator for cloud parameters developed in the CLOUDS Laboratory, at the Computer Science and Software Engineering Department of the University of Melbourne.

The CloudSim library is used for the following operations:

- Large scale cloud computing at data centres
- Virtualised server hosts with customisable policies
- Support for modelling and simulation of large scale cloud computing data centres
- Support for modelling and simulation of virtualised server hosts, with customisable policies for provisioning host resources to VMs
- Support for modelling and simulation of energy-aware computational resources
- Support for modelling and simulation of data centre network topologies and message-passing applications
- Support for modelling and simulation of federated clouds
- Support for dynamic insertion of simulation elements, as well as stopping and resuming simulation
- Support for user-defined policies to allot hosts to VMs, and policies for allotting host resources to VMs
- User-defined policies for allocation of hosts to virtual machines